

AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

**Dept. of Computer Science
Faculty of Science and Technology**

CSC2210: OBJECT ORIENTED PROGRAMMING 2

Summer 2025-2026

Section: D

Group No: 08

Project Report On

SkillBazaar — Online Course/Tuition Marketplace

Supervised By

Dr. Md. Iftekharul Mobin

Name	ID
Samorpita Paul	24-59060-3
Redwan Ahmed Chowdhury	24-56819-1
Sadia Islam Shorna	23-51988-2
Saurav Avik Biswas	25-60982-1

Obtained Marks for CO2 and CO3 (Description given below)

CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.

Evaluation Criteria	0	1-2	3	4	5
Requirement fulfillment					
Validation					
Verification					

CO3: Prepare and Explain a real life desktop based application synthesizing several component of C# along with development tools to adhere the given requirements.

Evaluation Criteria	0	1-2	3	4	5
Organization of the application					
Representation and Integration of Database					
Graphical User Interface					

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Chapter 1: Introduction & Case Study

Bangladesh has seen rapid growth in demand for supplementary education — university students preparing for admission tests, professionals learning new technical skills, and school students seeking subject tutoring. Currently, independent instructors and small coaching centers rely on scattered channels — Facebook pages, WhatsApp groups, or word-of-mouth — to sell their courses, with no central platform to manage payments, track enrollment, or build credibility through reviews.

SkillBazaar is a three-tier online marketplace that solves this by connecting independent instructors and coaching centers with students, while the platform owner (Super Admin) earns a commission on every transaction. It follows a “one platform, many sellers” model similar to Daraz or Foodpanda, but applied to online education. In this system, there are three types of users — Super Admin, Instructor (Admin), and Student (Customer). A Super Admin can approve or suspend instructors and moderate the platform. An Instructor can create, edit, and manage their own courses, and view their earnings. A Student can browse courses, add them to a cart, check out, and rate courses after purchase.

How it works: Instructors register on the platform and submit their institute details for approval by the Super Admin — a required step so the platform maintains quality and prevents scam listings. Once approved, an instructor can list unlimited courses, setting the price, category, available seats, and whether the course is a one-time purchase or a recurring subscription. Students browse the catalog, filter by category or price, add courses to a cart, and check out using mobile financial services (bKash/Nagad) or card payment. On every successful payment, the system automatically splits the amount — a fixed 20% commission is credited to the platform's revenue, and the remaining 80% is added to the instructor's pending balance. Instructors can request a payout of their accumulated balance at any time, which the Super Admin reviews and approves.

Beyond transactions, the platform builds trust through a review system — students who complete a course can rate and comment on it, helping future students choose quality courses and giving the Super Admin visibility into underperforming instructors. The Super Admin also monitors platform-wide metrics: total revenue, best-selling courses, and top-earning instructors.

Chapter 2: Functional Requirements & User Stories

2.1 Super Admin

- FR1. Super Admin can log in through a dedicated portal.
- FR2. Super Admin can view all pending instructor registrations and approve or reject them.
- FR3. Super Admin can suspend or reactivate any instructor account.
- FR4. Super Admin can view/delete any course that violates platform policy.
- FR5. Super Admin can view platform-wide reports: total revenue, best-selling courses, top-earning instructors.
- FR6. Super Admin can view and approve/reject instructor payout requests.
- FR7. Super Admin can view courses/instructors with an average rating below a threshold (e.g. 2.5).

User Stories:

- As a Super Admin, I want to review pending instructor applications so that only legitimate instructors can sell courses.
- As a Super Admin, I want to see total platform revenue so that I can track business performance.
- As a Super Admin, I want to suspend an instructor with repeated bad reviews so that the platform maintains quality.
- As a Super Admin, I want to approve instructor payout requests so that instructors receive their earnings.

2.2 Instructor (Admin)

- FR8. Instructor can register and submit institute details for approval.
- FR9. Instructor can log in only after Super Admin approval.
- FR10. Instructor can create, edit, and delete their own courses.
- FR11. Instructor can set course price, seat count, category, and pricing type (one-time/subscription).
- FR12. Instructor can view a list of students enrolled in their courses.
- FR13. Instructor can view their current earnings balance (after 20% platform commission).
- FR14. Instructor can request a payout of their balance.
- FR15. Instructor can view ratings/reviews left on their courses.

User Stories:

- As an Instructor, I want to create a new course listing so that students can find and buy it.
- As an Instructor, I want to see how many seats are left so that I don't oversell a course.
- As an Instructor, I want to view my earnings balance so that I know how much I can withdraw.
- As an Instructor, I want to request a payout so that I receive my money from course sales.

2.3 Customer (Student)

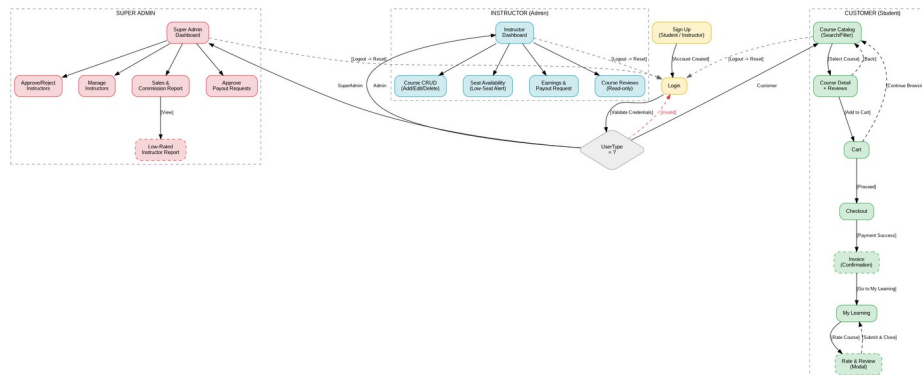
- FR16. Student can register and log in.
- FR17. Student can browse/search courses by title.
- FR18. Student can filter courses by category, price range, and rating.
- FR19. Student can add a course to their cart, update quantity, or remove it.
- FR20. Student can check out and pay via bKash/Nagad/Card (simulated).
- FR21. Student can view active discount offers applied at checkout.
- FR22. Student can view their enrolled/purchased courses (“My Learning”).
- FR23. Student can leave a rating and review for a course they've enrolled in.

User Stories:

- As a Student, I want to filter courses by price so that I can find one within my budget.
- As a Student, I want to add multiple courses to a cart so that I can check out together.
- As a Student, I want to pay with bKash so that I can complete my purchase easily.
- As a Student, I want to leave a review after finishing a course so that I can help other students decide.
- As a Student, I want to see my enrolled courses in one place so that I can access them anytime.

Chapter 3: UI Navigation Diagram

The diagram below shows every form in the SkillBazaar application and every arrow between them, starting from a shared Sign Up / Login entry point. After login, a decision node checks UserType and routes to one of three colour-coded dashboards. Every child form reachable from each dashboard is shown, with labelled arrows describing the triggering action (e.g. [Add to Cart], [Payment Success], [Logout → Reset]). Modal or transient states — Invoice confirmation and the Rate & Review popup — are drawn with a dashed border to distinguish them from full navigational forms.

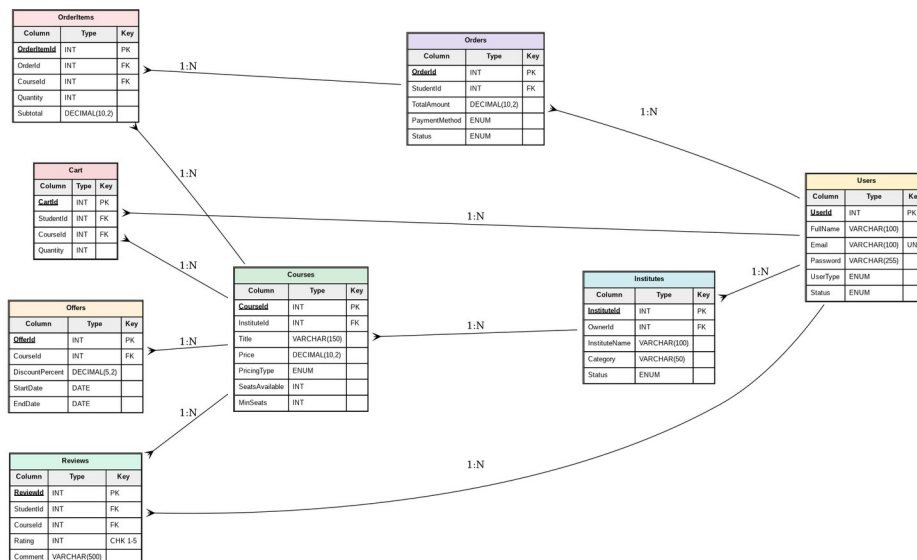


Flow explanation: A new user starts at Sign Up, then logs in. The Login form validates credentials against the Users table and reads UserType to decide the landing dashboard. Super Admins land on moderation and reporting tools. Instructors land on course management tools. Customers land on the course catalog, and can move forward through Course Detail → Cart → Checkout → Invoice → My Learning → Rate & Review, with Back/Continue-Browsing arrows allowing backward navigation at each step. Every dashboard has a Logout arrow that resets the session back to the Login form.

Chapter 4: Database Design

4.1 SQL Schema Diagram

The diagram below shows every table, its columns, data types, and primary/foreign key relationships.



4.2 Table Descriptions

Users

Column	Data Type	Constraint	Description
UserId	INT	PK, Auto Increment	Unique identifier for every user
FullName	VARCHAR(100)	NOT NULL	User's full name
Email	VARCHAR(100)	UNIQUE, NOT NULL	Login identifier
Password	VARCHAR(255)	NOT NULL	Hashed password
UserType	ENUM	SuperAdmin/Admin/Customer	Determines role-based routing
Status	ENUM	Pending/Approved/Suspended	Account moderation state

Institutes

Column	Data Type	Constraint	Description
InstituteId	INT	PK, Auto Increment	Unique identifier
OwnerId	INT	FK → Users(UserId)	Instructor who owns this institute
InstituteName	VARCHAR(100)	NOT NULL	Display name
Category	VARCHAR(50)	NOT NULL	Subject area
Status	ENUM	Pending/Approved/Suspended	Approval state set by Super Admin

Courses

Column	Data Type	Constraint	Description
CourseId	INT	PK, Auto Increment	Unique identifier

Column	Data Type	Constraint	Description
InstituteId	INT	FK → Institutes	Owning institute
Title	VARCHAR(150)	NOT NULL	Course name
Price	DECIMAL(10,2)	CHECK (Price >= 0)	Listed price
PricingType	ENUM	OneTime/Subscription	Billing model
SeatsAvailable	INT	DEFAULT 0	Remaining capacity
MinSeats	INT	DEFAULT 5	Low-seat alert threshold

Cart, Orders & OrderItems

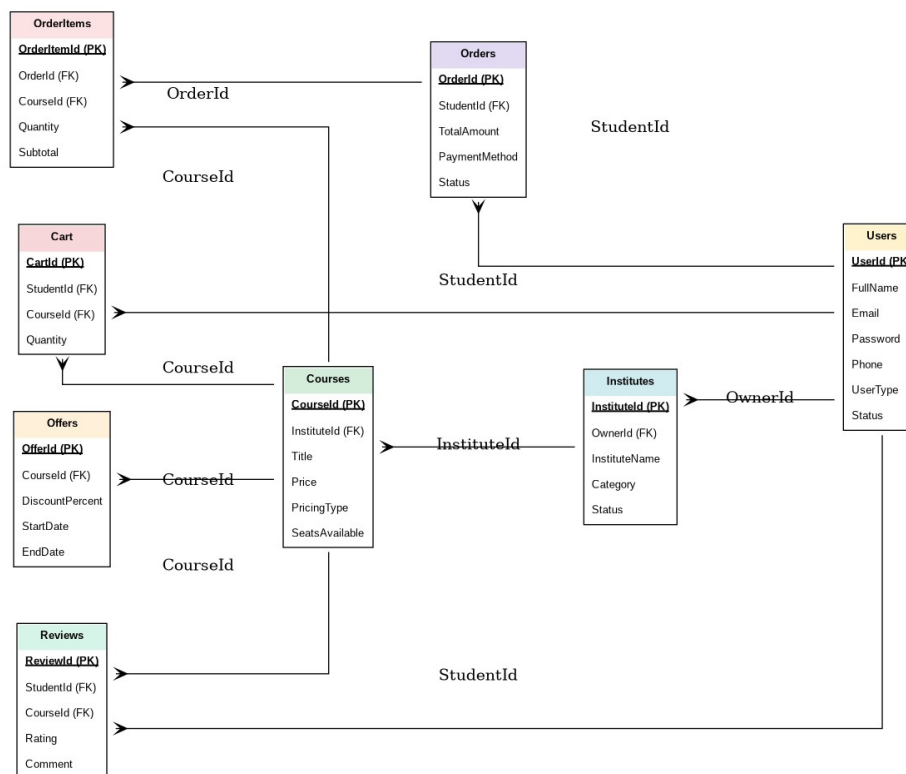
Column	Data Type	Constraint	Description
CartId	INT	PK, Auto Increment	Basket line item
OrderId	INT	PK, Auto Increment	One row per checkout
TotalAmount	DECIMAL(10,2)	CHECK (>= 0)	Sum paid
PaymentMethod	ENUM	bKash/Nagad/Card/Cash	Payment channel
OrderItemId	INT	PK, Auto Increment	Junction table row

Note: OrderItems is a mandatory junction table that resolves the many-to-many relationship between Orders and Courses — each order can contain multiple courses, and each course can appear in multiple orders. Storing a comma-separated course list inside Orders directly would violate 1NF.

Reviews & Offers

Column	Data Type	Constraint	Description
ReviewId	INT	PK, Auto Increment	Unique identifier
Rating	INT	CHECK (1-5)	Star rating
OfferId	INT	PK, Auto Increment	Unique identifier
DiscountPercent	DECIMAL(5,2)	CHECK (0-100)	Percentage off

4.3 Entity-Relationship Diagram



4.4 Normalisation Justification (3NF)

The schema satisfies Third Normal Form:

- 1NF: every column holds a single atomic value — for example, an order's courses are not stored as a comma-separated list but broken into separate OrderItems rows.
- 2NF: every non-key column depends on the whole primary key. Each table uses a single-column surrogate key (e.g. CourseId), so partial dependency cannot occur.
- 3NF: no non-key column depends on another non-key column. For example, InstituteName is stored only in Institutes, not repeated in Courses — Courses only stores the InstituteId foreign key, avoiding transitive dependency.
- The many-to-many relationship between Orders and Courses is correctly resolved with the OrderItems junction table rather than a repeating group.

Chapter 5: SQL Queries

Below are the SQL statements grouped by feature, matching the functional requirements defined in Chapter 2. The complete script is also available at database/schema.sql in the GitHub repository.

5.1 Database & Table Creation

Creates the database and the core Users table that stores all three roles.

```
CREATE DATABASE IF NOT EXISTS SkillBazaar;
USE SkillBazaar;

CREATE TABLE Users (
  UserId      INT AUTO_INCREMENT PRIMARY KEY,
  FullName    VARCHAR(100) NOT NULL,
  Email       VARCHAR(100) NOT NULL UNIQUE,
  Password    VARCHAR(255) NOT NULL,
  Phone       VARCHAR(20),
  Address     VARCHAR(200),
  UserType    ENUM('SuperAdmin','Admin','Customer') NOT NULL,
  Status      ENUM('Pending','Approved','Suspended') NOT NULL DEFAULT 'Approved',
  CreatedAt   DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
);
```

5.2 User Registration & Authentication

Sign Up (Registration)

```
INSERT INTO Users (FullName, Email, Password, Phone, Address, UserType, Status)
VALUES ('Samorpita Islam', 'samorpita@student.com', 'hashed_pw_4',
       '01700000004', 'Mirpur, Dhaka', 'Customer', 'Approved');
```

Sign In (Login) — verifies credentials and returns role for landing-page routing

```
SELECT UserId, FullName, UserType
FROM Users
WHERE Email = 'samorpita@student.com' AND Password = 'hashed_pw_4';
```

Update Password

```
UPDATE Users SET Password = 'newHashedPassword123' WHERE UserId = 4;
```

5.3 Course Browsing, Filtering & Search

Filter by price range

```
SELECT * FROM Courses WHERE Price BETWEEN 1000 AND 2500;
```

Filter by category and availability (seats > 0)

```
SELECT * FROM Courses WHERE Category = 'Programming' AND SeatsAvailable > 0;
```

Search by keyword

```
SELECT * FROM Courses WHERE Title LIKE '%web%' OR Description LIKE '%web%';
```

5.4 Shopping Cart & Checkout

Add item to cart

```
INSERT INTO Cart (StudentId, CourseId, Quantity) VALUES (4, 2, 1);
```

Delete item from cart

```
DELETE FROM Cart WHERE CartId = 1;
```

View cart with line totals

```
SELECT c.CartId, co.Title, co.Price, c.Quantity, (co.Price * c.Quantity) AS LineTotal
FROM Cart c JOIN Courses co ON c.CourseId = co.CourseId
WHERE c.StudentId = 4;
```

Checkout: insert Order + OrderItems + decrement seats

```
INSERT INTO Orders (StudentId, TotalAmount, PaymentMethod, Status)
VALUES (4, 1500.00, 'bKash', 'Paid');
```

```
INSERT INTO OrderItems (OrderId, CourseId, Quantity, UnitPrice, Subtotal)
VALUES (LAST_INSERT_ID(), 1, 1, 1500.00, 1500.00);
```

```
UPDATE Courses SET SeatsAvailable = SeatsAvailable - 1 WHERE CourseId = 1;
```

5.5 Instructor Earnings (JOIN + GROUP BY + SUM)

```
SELECT i.InstituteName AS Instructor,
       SUM(oi.Subtotal) * 0.80 AS NetEarnings,
       SUM(oi.Subtotal) * 0.20 AS PlatformCommission
FROM OrderItems oi
JOIN Courses c ON oi.CourseId = c.CourseId
JOIN Institutes i ON c.InstituteId = i.InstituteId
GROUP BY i.InstituteName;
```

5.6 Low Seat Alert

```
SELECT Title, SeatsAvailable, MinSeats FROM Courses WHERE SeatsAvailable < MinSeats;
```

5.7 Reviews (JOIN)

```
SELECT u.FullName, r.Rating, r.Comment, r.ReviewDate
FROM Reviews r JOIN Users u ON r.StudentId = u.UserId
WHERE r.CourseId = 1;
```

5.8 Super Admin: Low-Rated Instructors (JOIN + GROUP BY + HAVING)

```
SELECT i.InstituteName, AVG(r.Rating) AS AvgRating
FROM Reviews r
JOIN Courses c ON r.CourseId = c.CourseId
JOIN Institutes i ON c.InstituteId = i.InstituteId
GROUP BY i.InstituteName
HAVING AVG(r.Rating) < 2.5;
```

5.9 Offers & Discounts

```
SELECT c.Title, c.Price, o.DiscountPercent,
       ROUND(c.Price * (1 - o.DiscountPercent / 100), 2) AS DiscountedPrice
FROM Offers o JOIN Courses c ON o.CourseId = c.CourseId
WHERE CURDATE() BETWEEN o.StartDate AND o.EndDate;
```

Full SQL Script Reference

The complete script — all CREATE TABLE statements, sample data (minimum 3 rows per table, including 1 Super Admin, 2 Institutes/Admins, and 3 Students), and all feature queries — is reproduced below.

```

-- =====
-- SkillBazaar : Online Course / Tuition Marketplace
-- CSC 2210 - Object Oriented Programming 2
-- Database: MySQL 8.0
-- =====

CREATE DATABASE IF NOT EXISTS SkillBazaar;
USE SkillBazaar;

-- =====
-- 1. TABLE CREATION
-- =====

-- Users: stores all three roles (SuperAdmin, Admin/Instructor, Customer/Student)
CREATE TABLE Users (
    UserId          INT AUTO_INCREMENT PRIMARY KEY,
    FullName        VARCHAR(100) NOT NULL,
    Email           VARCHAR(100) NOT NULL UNIQUE,
    Password        VARCHAR(255) NOT NULL,
    Phone           VARCHAR(20),
    Address          VARCHAR(200),
    UserType        ENUM('SuperAdmin','Admin','Customer') NOT NULL,
    Status          ENUM('Pending','Approved','Suspended') NOT NULL DEFAULT 'Approved',
    CreatedAt       DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
);

-- Institutes: one row per Admin (instructor / coaching center / course provider)
CREATE TABLE Institutes (
    InstituteId     INT AUTO_INCREMENT PRIMARY KEY,
    OwnerId         INT NOT NULL,
    InstituteName   VARCHAR(100) NOT NULL,
    Category        VARCHAR(50) NOT NULL,
    Address         VARCHAR(200),
    ContactPhone    VARCHAR(20),
    Status          ENUM('Pending','Approved','Suspended') NOT NULL DEFAULT 'Pending',
    FOREIGN KEY (OwnerId) REFERENCES Users(UserId)
);

-- Courses: items for sale, owned by exactly one Institute
CREATE TABLE Courses (
    CourseId        INT AUTO_INCREMENT PRIMARY KEY,
    InstituteId     INT NOT NULL,
    Title           VARCHAR(150) NOT NULL,
    Category        VARCHAR(50) NOT NULL,
    Price           DECIMAL(10,2) NOT NULL CHECK (Price >= 0),
    PricingType     ENUM('OneTime','Subscription') NOT NULL DEFAULT 'OneTime',
    DurationMonths  INT NULL, -- used only when PricingType = 'Subscription'
    SeatsAvailable  INT NOT NULL DEFAULT 0 CHECK (SeatsAvailable >= 0),
    MinSeats        INT NOT NULL DEFAULT 5, -- low-availability alert threshold
    Description     VARCHAR(1000),
    ImagePath       VARCHAR(255),
    FOREIGN KEY (InstituteId) REFERENCES Institutes(InstituteId)
);

-- Cart: a student's active basket before checkout
CREATE TABLE Cart (
    CartId          INT AUTO_INCREMENT PRIMARY KEY,
    StudentId       INT NOT NULL,
    CourseId        INT NOT NULL,
    Quantity        INT NOT NULL DEFAULT 1 CHECK (Quantity > 0),
    AddedDate       DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (StudentId) REFERENCES Users(UserId),
    FOREIGN KEY (CourseId) REFERENCES Courses(CourseId)
);

-- Orders: one row per completed purchase/enrollment
CREATE TABLE Orders (
    OrderId         INT AUTO_INCREMENT PRIMARY KEY,
    StudentId       INT NOT NULL,

```

```

    OrderDate      DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    TotalAmount    DECIMAL(10,2) NOT NULL CHECK (TotalAmount >= 0),
    PaymentMethod  ENUM('bKash','Nagad','Card','Cash') NOT NULL,
    Status         ENUM('Pending','Paid','Cancelled') NOT NULL DEFAULT 'Pending',
    FOREIGN KEY (StudentId) REFERENCES Users(UserId)
);

-- OrderItems: junction table -- an order has many courses, a course appears in many orders
CREATE TABLE OrderItems (
    OrderItemId    INT AUTO_INCREMENT PRIMARY KEY,
    OrderId        INT NOT NULL,
    CourseId       INT NOT NULL,
    Quantity       INT NOT NULL DEFAULT 1 CHECK (Quantity > 0),
    UnitPrice      DECIMAL(10,2) NOT NULL,
    Subtotal       DECIMAL(10,2) NOT NULL,
    FOREIGN KEY (OrderId) REFERENCES Orders(OrderId),
    FOREIGN KEY (CourseId) REFERENCES Courses(CourseId)
);

-- Reviews: ratings & comments left by students on courses
CREATE TABLE Reviews (
    ReviewId       INT AUTO_INCREMENT PRIMARY KEY,
    StudentId      INT NOT NULL,
    CourseId       INT NOT NULL,
    Rating         INT NOT NULL CHECK (Rating BETWEEN 1 AND 5),
    Comment        VARCHAR(500),
    ReviewDate     DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (StudentId) REFERENCES Users(UserId),
    FOREIGN KEY (CourseId) REFERENCES Courses(CourseId)
);

-- Offers: discount campaigns on a course for a date range
CREATE TABLE Offers (
    OfferId        INT AUTO_INCREMENT PRIMARY KEY,
    CourseId       INT NOT NULL,
    DiscountPercent DECIMAL(5,2) NOT NULL CHECK (DiscountPercent BETWEEN 0 AND 100),
    StartDate      DATE NOT NULL,
    EndDate        DATE NOT NULL,
    FOREIGN KEY (CourseId) REFERENCES Courses(CourseId)
);

-- =====
-- 2. SAMPLE DATA (>=3 rows per table: 1 SuperAdmin, 2 Institutes/Admins, 3 Students)
-- =====

INSERT INTO Users (FullName, Email, Password, Phone, Address, UserType, Status) VALUES
('Platform Owner', 'owner@skillbazaar.com', 'hashed_pw_1', '01700000001', 'Dhaka, Bangladesh', 'SuperAdmin', 'Approved'),
('Rafiq Ahmed', 'rafiq@codersbd.com', 'hashed_pw_2', '01700000002', 'Uttara, Dhaka', 'Admin', 'Approved'),
('Nusrat Jahan', 'nusrat@examprep.com', 'hashed_pw_3', '01700000003', 'Dhanmondi, Dhaka', 'Admin', 'Pending'),
('Samorpita Islam', 'samorpita@student.com', 'hashed_pw_4', '01700000004', 'Mirpur, Dhaka', 'Customer', 'Approved'),
('Tosim Hossain', 'tosim@student.com', 'hashed_pw_5', '01700000005', 'Banani, Dhaka', 'Customer', 'Approved'),
('Fahim Rahman', 'fahim@student.com', 'hashed_pw_6', '01700000006', 'Mohammadpur, Dhaka', 'Customer', 'Approved');

INSERT INTO Institutes (OwnerId, InstituteName, Category, Address, ContactPhone, Status) VALUES
(2, 'Coders BD Academy', 'Programming', 'Uttara, Dhaka', '01700000002', 'Approved'),
(3, 'ExamPrep Institute', 'Admission Prep', 'Dhanmondi, Dhaka', '01700000003', 'Pending');

INSERT INTO Courses (InstituteId, Title, Category, Price, PricingType, DurationMonths, SeatsAvailable, MinSeats, Description, ImagePath) VALUES
(1, 'C# OOP Fundamentals', 'Programming', 1500.00, 'OneTime', NULL, 40, 10, 'Learn classes, inheritance, polymorphism in C#.', 'img/csharp.png'),
(1, 'Full Stack Web Development', 'Programming', 3000.00, 'Subscription', 6, 25, 5, 'HTML,
```

```

CSS, JS, React and Node in 6 months.', 'img/webdev.png'),
(2, 'AIUB Admission Crash Course', 'Admission Prep', 2000.00, 'OneTime', NULL, 60, 15,
'Complete prep for AIUB admission test.', 'img/admission.png');

INSERT INTO Cart (StudentId, CourseId, Quantity) VALUES
(4, 1, 1),
(5, 3, 1),
(6, 2, 1);

INSERT INTO Orders (StudentId, OrderDate, TotalAmount, PaymentMethod, Status) VALUES
(4, '2026-08-01 10:15:00', 1500.00, 'bKash', 'Paid'),
(5, '2026-08-05 14:30:00', 2000.00, 'Card', 'Paid'),
(6, '2026-08-10 09:00:00', 3000.00, 'Nagad', 'Pending');

INSERT INTO OrderItems (OrderId, CourseId, Quantity, UnitPrice, Subtotal) VALUES
(1, 1, 1, 1500.00, 1500.00),
(2, 3, 1, 2000.00, 2000.00),
(3, 2, 1, 3000.00, 3000.00);

INSERT INTO Reviews (StudentId, CourseId, Rating, Comment, ReviewDate) VALUES
(4, 1, 5, 'Excellent explanation of OOP concepts, very practical.', '2026-08-03 12:00:00'),
(5, 3, 2, 'Content was too basic for the price.', '2026-08-06 18:20:00'),
(6, 2, 4, 'Good course but the pace is fast.', '2026-08-11 20:10:00');

INSERT INTO Offers (CourseId, DiscountPercent, StartDate, EndDate) VALUES
(1, 20.00, '2026-08-10', '2026-08-31'),
(3, 15.00, '2026-08-01', '2026-08-20');

-- =====
-- 3. FEATURE QUERIES (one per major feature)
-- =====

-- 1. LOGIN: verify credentials and get role for landing-page routing
SELECT UserId, FullName, UserType
FROM Users
WHERE Email = 'samorpita@student.com' AND Password = 'hashed_pw_4';

-- 2. FILTER: courses by price range
SELECT * FROM Courses
WHERE Price BETWEEN 1000 AND 2500;

-- 3. FILTER: courses by category + availability (seats > 0)
SELECT * FROM Courses
WHERE Category = 'Programming' AND SeatsAvailable > 0;

-- 4. SEARCH: courses by keyword
SELECT * FROM Courses
WHERE Title LIKE '%web%' OR Description LIKE '%web%';

-- 5a. CART: add item
INSERT INTO Cart (StudentId, CourseId, Quantity) VALUES (4, 2, 1);

-- 5b. CART: delete item
DELETE FROM Cart WHERE CartId = 1;

-- 5c. CART: view cart with total
SELECT c.CartId, co.Title, co.Price, c.Quantity, (co.Price * c.Quantity) AS LineTotal
FROM Cart c
JOIN Courses co ON c.CourseId = co.CourseId
WHERE c.StudentId = 4;

-- 6. CHECKOUT: insert Order + insert OrderItems + decrement seats
-- (run as a sequence / transaction from the application layer)
INSERT INTO Orders (StudentId, TotalAmount, PaymentMethod, Status)
VALUES (4, 1500.00, 'bKash', 'Paid');

INSERT INTO OrderItems (OrderId, CourseId, Quantity, UnitPrice, Subtotal)
VALUES (LAST_INSERT_ID(), 1, 1, 1500.00, 1500.00);

```

```

UPDATE Courses SET SeatsAvailable = SeatsAvailable - 1 WHERE CourseId = 1;

-- 7. INSTITUTE EARNINGS: total sales grouped by institute (JOIN + GROUP BY + SUM)
SELECT i.InstituteName, SUM(oi.Subtotal) AS TotalEarnings
FROM OrderItems oi
JOIN Courses c ON oi.CourseId = c.CourseId
JOIN Institutes i ON c.InstituteId = i.InstituteId
GROUP BY i.InstituteName;

-- 8. LOW SEAT ALERT: courses where SeatsAvailable < MinSeats
SELECT Title, SeatsAvailable, MinSeats
FROM Courses
WHERE SeatsAvailable < MinSeats;

-- 9. REVIEWS: all reviews for one course with reviewer name (JOIN)
SELECT u.FullName, r.Rating, r.Comment, r.ReviewDate
FROM Reviews r
JOIN Users u ON r.StudentId = u.UserId
WHERE r.CourseId = 1;

-- 10. SUPER ADMIN: institutes with average rating < 2.5 (JOIN + GROUP BY + HAVING)
SELECT i.InstituteName, AVG(r.Rating) AS AvgRating
FROM Reviews r
JOIN Courses c ON r.CourseId = c.CourseId
JOIN Institutes i ON c.InstituteId = i.InstituteId
GROUP BY i.InstituteName
HAVING AVG(r.Rating) < 2.5;

-- 11. SUPER ADMIN: suspend/delete an institute owner
UPDATE Institutes SET Status = 'Suspended' WHERE InstituteId = 2;

-- 12. OFFERS: active offers today with calculated discounted price
SELECT c.Title, c.Price, o.DiscountPercent,
       ROUND(c.Price * (1 - o.DiscountPercent/100), 2) AS DiscountedPrice
FROM Offers o
JOIN Courses c ON o.CourseId = c.CourseId
WHERE CURDATE() BETWEEN o.StartDate AND o.EndDate;

```

Chapter 6: User Interface Design

All 12 required form mockups are shown below, each with a caption. A consistent colour scheme (purple for navigation, green for student actions, blue for instructor actions, red for admin actions) and button style is used across every form. Data lists use table/DataGridView-style layouts rather than plain text lists, and filters use dropdowns rather than free-text boxes. Validation states are shown in the Sign Up mockup (invalid email, short password).

6.1 Login

Screen 1: Login / Register

SkillBazaar

Learn from the best. Sell what you know.

Email

Password

[Log In](#)

Don't have an account? [Sign up as Student](#) or [Sign up as Instructor](#)

Login

6.2 Sign Up / Registration

Screen: Sign Up / Registration

Create Your Account

☒ I'm a Student
 ☐ I'm an Instructor

Full Name

Samorpita Islam

Email

samorpita@

△ Enter a valid email address

Password

12345

△ Password must be at least 6 characters

Phone

01700000004

Create Account

Already have an account? [Log In](#)

Sign Up / Registration

6.3 Super Admin Dashboard

Screen 5: Super Admin Dashboard

Super Admin

- Overview
- Instructor Approvals
- All Courses
- Revenue Reports
- Payout Requests

Platform Overview

₹96,400
 Platform Revenue (20% commission)

18
 Active Instructors

3
 Pending Approvals

Pending Instructor Approvals

Institute	Owner	Category	Action
ExamPrep Institute	Nusrat Jahan	Admission Prep	Approve Reject

Super Admin Dashboard

6.4 Super Admin — Manage Instructors

Screen: Super Admin — Manage Instructors (Approve/Suspend)

Super Admin

- Overview
- Instructor Approvals
- Manage Instructors**
- Revenue Reports
- Payout Requests

All Instructors

Status: All ▼
 Category: All ▼

Institute	Owner	Category	Status	Action
Coders BD Academy	Rafiq Ahmed	Programming	Approved	Suspend
ExamPrep Institute	Nusrat Jahan	Admission Prep	Pending	Approve Reject
MathGuru BD	Kamal Hossain	Mathematics	Approved	Suspend

Super Admin — Manage Instructors

6.5 Super Admin — Platform Sales & Commission Report

Screen: Super Admin — Platform Sales & Commission Report

Super Admin

- Overview
- Instructor Approvals
- Manage Instructors
- Revenue Reports
- Payout Requests

Platform Sales & Commission Report

₹482,000
Total Sales (all instructors)

₹96,400
Platform Commission (20%)

227
Total Enrollments

Instructor	Courses Sold	Gross Sales	Platform Commission	Instructor Net
Coders BD Academy	142	₹241,000	₹48,200	₹192,800
ExamPrep Institute	60	₹120,000	₹24,000	₹96,000
MathGuru BD	25	₹121,000	₹24,200	₹96,800

Super Admin — Platform Sales & Commission Report

6.6 Instructor — Course CRUD

Screen: Instructor — Course CRUD (Add/Update/Delete)

Coders BD Academy

- Dashboard
- My Courses
- Students
- Earnings & Payout
- Reviews

My Courses

+ Add New Course

ID	Title	Category	Price	Seats	Action
1	C# OOP Fundamentals	Programming	₹1500	39/40	Edit Delete
2	Full Stack Web Dev	Programming	₹3000	24/25	Edit Delete

Course Title

Price (₹)

Seats Available

Save Course

Instructor — Course CRUD

6.7 Instructor — Seat Availability Dashboard

Screen: Instructor — Seat Availability Dashboard (Low-Seat Alert)

Coders BD Academy

- Dashboard
- My Courses
- Seat Availability
- Earnings & Payout
- Reviews

Seat Availability

Course	Total Seats	Available	Min Threshold	Status
C# OOP Fundamentals	40	39	10	OK
Full Stack Web Dev	25	3	5	⚠️ LOW SEATS
Data Structures	30	28	5	OK

⚠️ 1 course is below its minimum seat threshold. Consider increasing capacity.

Instructor — Seat Availability Dashboard

6.8 Instructor — Earnings Dashboard

Screen 4: Instructor Dashboard

Coders BD Academy

- Dashboard
- My Courses
- Students
- Earnings & Payout
- Reviews

Welcome back, Rafiq

142
Total Enrollments

₹48,200
Available Balance (after 20% fee)

4.4 ★
Avg. Rating

[+ Add New Course](#)
[Request Payout](#)

Course	Seats Left	Price	Enrolled
C# OOP Fundamentals	39/40	₹1500	1
Full Stack Web Dev	24/25	₹3000	1

Instructor — Earnings Dashboard

6.9 Customer — Course Catalog (Search & Filter)

Screen 2: Student — Course Catalog

SkillBazaar

Catalog Cart (2) My Learning Profile

Category: Programming ▼

Price: ₹0-5000 ▼

Rating: 4+ ▼

C# OOP Fundamentals
Coders BD Academy · ★ 4.6
₹1500 [Add to Cart](#)

Full Stack Web Dev
Coders BD Academy · ★ 4.2
₹3000 /6mo [Add to Cart](#)

AIUB Admission Prep
ExamPrep Institute · ★ 3.8
₹2000 [Add to Cart](#)

Customer — Course Catalog (Search & Filter)

6.10 Customer — Course Details + Reviews

Screen: Course Details + Reviews

SkillBazaar

Catalog Cart (2) My Learning

C# OOP Fundamentals

by Coders BD Academy

★★★★★ 4.6 (38 reviews)

₹1500

Learn classes, inheritance, polymorphism, and interfaces in C# through hands-on projects.

[Add to Cart](#)

Student Reviews

Samorpita Islam ★★★★★

Excellent explanation of OOP concepts, very practical.

Tosim Hossain ★★★★★

Good course but the pace is fast.

Customer — Course Details + Reviews

6.11 Customer — Cart

Screen: Cart

SkillBazaar [Catalog](#) **Cart (2)** [My Learning](#)

Your Cart

Course	Instructor	Price	Qty	Subtotal	
C# OOP Fundamentals	Coders BD Academy	₳1500	- 1 +	₳1500	Remove
AIUB Admission Prep	ExamPrep Institute	₳2000	- 1 +	₳2000	Remove

Cart Total: ₳3500

[Continue Browsing](#) [Proceed to Checkout →](#)

Customer — Cart

6.12 Customer — Checkout / Invoice

Screen: Checkout & Invoice

SkillBazaar [Catalog](#) [Cart](#) [My Learning](#)

Checkout

Payment Method

☒ bKash ☐ Nagad ☐ Card

Active Offer Applied

AIUB Admission Prep — 15% OFF (Offer ends Aug 20)

[Confirm & Pay ₳3200](#)

✓ **Payment Successful — Invoice #ORD-1042**

Course	Instructor	Price Paid
C# OOP Fundamentals	Coders BD Academy	₳1500
AIUB Admission Prep	ExamPrep Institute	₳1700

Total Paid: ₳3200 via bKash on Aug 17, 2026

[Download Invoice \(PDF\)](#) [Go to My Learning](#)

Customer — Checkout / Invoice

Chapter 7: Work Distribution

Name	ID	Contribution
Samorpita Paul	24-59060-3	28% — Database schema design (schema.sql, ER diagram, SQL schema diagram), UI navigation diagram, all 12 UI mockups
Sadia Islam Shorna	23-51988-2	24% — Case study, functional requirements & user stories
Redwan Ahmed Chowdhury	24-56819-1	24% — SQL queries (Chapter 5), work distribution & conclusion
Saurav Avik Biswas	25-60982-1	24% — Report compilation, README.md

Chapter 8: Conclusion & Future Work

SkillBazaar demonstrates a complete three-tier marketplace design connecting Super Admin, Instructor, and Student roles through a shared database and a consistent navigation flow. The design satisfies the core Object-Oriented Programming project requirements: normalized 3NF database with a proper junction table for many-to-many order/course relationships, role-based access control, and a commission-based revenue model that gives every stakeholder a reason to use the platform.

Future Work — in the coding phase of this project, the following will be implemented:

- Build the actual C# WinForms application implementing every form shown in Chapter 6, connected to a live MySQL database via a dedicated Database Connection class.
- Implement full CRUD operations for courses, cart, and orders using parameterized queries to prevent SQL injection.
- Implement proper OOP structure: an abstract/base User class with Student, Instructor, and SuperAdmin subclasses using inheritance and polymorphism (e.g. overriding a GetDashboard() method per role).
- Add real payment gateway integration (or a more realistic simulation) in place of the current simulated bKash/Nagad/Card flow.
- Add form-level validation and exception handling throughout, matching the states shown in the Sign Up mockup.
- Test the low-seat alert and payout-approval workflows end-to-end.

Challenges expected: correctly synchronizing seat counts during concurrent checkouts, safely handling the commission-split calculation without rounding errors, and ensuring the WinForms UI stays responsive during database calls.